

3D Multi-View Stereo Reconstruction: ETH3D and Middlebury Chess

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Abstract—This report presents a comprehensive pipeline for multi-view 3D reconstruction using both the ETH3D and Middlebury Chess (Mobile 2021) datasets. We detail all major steps, including stereo calibration, rectification, block matching, disparity refinement, 3D point cloud generation, and multi-view fusion. Each stage is critically analyzed: we discuss parameter optimization, technical challenges, and quantitative evaluation at both the pixel and point-cloud levels. Our report demonstrates the importance of robust calibration, careful parameter selection, and the practical difficulties of fusing multi-view reconstructions when ground truth extrinsic matrices are unavailable.

Keywords—*multi-view stereo, ETH3D, Middlebury, disparity, point cloud, ICP fusion, ball-pivoting, 3D mesh*

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1. Introduction

Dense 3D reconstruction from calibrated images remains a cornerstone problem in computer vision, with applications in robotics, augmented reality, and digital archiving. Achieving accurate reconstructions from stereo images poses several fundamental challenges, including camera geometry ambiguity, noisy depth estimation, and inconsistencies in multi-view fusion.

Despite advances in deep learning and end-to-end neural approaches, current gaps persist:

- **Single-view methods** often lack the geometric precision needed for robust 3D recovery.
- **End-to-end neural pipelines** require large annotated datasets and still struggle with real-world noise.
- **Real-world factors** such as low-texture regions, occlusions, and suboptimal image selection further degrade mesh quality and reconstruction accuracy.

1.1. Objective

This report systematically evaluates the importance of scene geometry and dataset choice in classical 3D reconstruction pipelines, with particular attention to:

- The impact of low-texture areas and occlusions on classical methods.
- How image pair selection and dataset properties (ETH3D vs. Middlebury Chess) influence final results.

1.2. Experimental Plan

- We conduct experiments at three levels: disparity map estimation, single-view point clouds, and multi-view fusion.
- Two datasets are studied: **ETH3D**, which is challenging and highlights the limitations of classical methods; and **Middlebury Chess (Mobile 2021)**, which allows us to clearly demonstrate the effects of our pipeline choices.
- Evaluation metrics include both **pixel-wise** (MAE, RMS, Bad1%, Bad2%, Bad4%) and **3D cloud-level criteria** (Chamfer distance, Accuracy, Completeness, F1 Score).

1.3. Individual Contributions

Anirban Das: Literature review and theoretical grounding; proposing experimental methodology; suggesting parameter choices and debugging when necessary.

Minh Khang Tran: Full coding pipeline implementation; proposing suitable datasets and quantitative evaluation metrics; generating visualizations and result reports for iterative review.

In summary, this report details an end-to-end pipeline for classical stereo and multi-view reconstruction, with careful technical analysis, parameter sensitivity studies, and a clear demonstration of practical limitations and solutions.

2. Related Work

Stereo and multi-view reconstruction have been intensively studied for decades. Traditional pipelines rely on accurate camera calibration, rectified stereo matching, and depth triangulation [hartley2004multiple](#), [szeliski2010computer](#). The ETH3D dataset [schoeps2017multi](#) and Middlebury benchmarks [scharstein2014high](#) are widely used for both classical and learning-based algorithms. State-of-the-art approaches include COLMAP for SfM and MVS [schonberger2016structure](#), OpenCV for SGBM block matching [opencv_doc](#), and various outlier removal/fusion algorithms [zhou2018open3d](#). Recent work highlights the persistent challenge of achieving robust, high-precision fusion, especially when extrinsic parameters are estimated rather than measured.

3. Experiments with the ETH3D Dataset

3.1. Pipeline Overview

Our pipeline follows a standard yet rigorous workflow:

1. **Load camera calibration and poses:** Intrinsics, extrinsics from COLMAP files.
2. **Select stereo pairs:** Based on baseline and overlap, balancing depth accuracy and texture matching.
3. **Rectify images:** Using the estimated relative pose, aligning epipolar lines horizontally.

4. **Preprocess images:** Apply CLAHE for contrast enhancement in low-texture regions.
5. **Compute disparity:** SGBM with left-right consistency check.
6. **Apply WLS filtering:** For smoothness and edge preservation.
7. **Reproject disparity to point cloud:** Using the rectification Q matrix.
8. **Clean and align** point clouds from multiple stereo pairs.
9. **Fuse and mesh** for a unified 3D model.

3.2. Data Setup and Image Selection

Camera intrinsics K (focal length, principal point) and extrinsics T_{wc} (4×4 world-to-camera matrices) are loaded using a COLMAP-compatible parser to extract calibration data including the sparse 3D point cloud. Baselines are computed as:

$$\text{baseline}_{i,j} = \|\mathbf{C}_i - \mathbf{C}_j\|_2, \quad \mathbf{C}_i = -R_i^T t_i \quad (1)$$

where R_i, t_i are from T_{wc} . Baselines guide the selection of stereo pairs: a balance is required (too small = little depth cue; too large = more occlusion, less overlap).

For our experiments, we selected three consecutive images (DSC_682, DSC_683, DSC_684) from the ETH3D high-resolution multi-view dataset's delivery area scene. This selection provides two overlapping stereo pairs with adequate baseline separation for reliable depth estimation.

3.3. Calibration and Rectification

Rectification is performed using OpenCV's `cv2.stereoRectify` to compute the rectification transformation, followed by `cv2.initUndistortRectifyMap` to generate rectification maps. The actual rectification is applied using `cv2.remap`, mapping both images so that corresponding points appear on the same horizontal line:

$$Q = \begin{bmatrix} 1 & 0 & 0 & -c_x \\ 0 & 1 & 0 & -c_y \\ 0 & 0 & 0 & f \\ 0 & 0 & 1/b & -d_{\text{offs}}/b \end{bmatrix} \quad (2)$$

where f is focal length, (c_x, c_y) is the principal point, b is baseline, d_{offs} is disparity offset.

3.4. Image Preprocessing

To improve stereo matching performance in low-texture regions, we apply Contrast Limited Adaptive Histogram Equalization (CLAHE) using `cv2.createCLAHE`. This adaptive normalization technique enhances local contrast, particularly beneficial for the large concrete wall surfaces present in the delivery area scene where texture information is limited.

3.5. Stereo Matching and Disparity Estimation

Disparity is estimated with Semi-Global Block Matching (SGBM), a cost-minimizing dynamic programming approach **opencv_doc**, enhanced with left-right consistency checking for improved reliability.

Key parameters and optimization:

- **minDisparity:** Set to 0 based on minimum scene depth analysis.
- **numDisparities:** Set to 192 to cover the full expected depth range in the delivery area scene.
- **blockSize:** Set to 3–5 to preserve fine details while maintaining matching robustness.
- **P1/P2:** Smoothness penalties ($P1=8 \times 3^2$, $P2=32 \times 3^2$), optimized to balance noise reduction and detail preservation.
- **uniquenessRatio:** Set to 10 to remove ambiguous matches.
- **WLS lambda, sigma:** Tuned empirically ($\lambda=12000.0$, $\sigma=2.0$) for minimum Chamfer distance on reprojected clouds.

The full SGBM+WLS pipeline:

```
sgbm = cv2.StereoSGBM_create(
    minDisparity=0, numDisparities=192, blockSize=3,
    P1=8*3**2, P2=32*3**2, disp12MaxDiff=2,
    uniquenessRatio=10,
    speckleWindowSize=100, speckleRange=2,
    mode=cv2.STEREO_SGBM_MODE_SGBM_3WAY
)
wls_filter = cv2.ximgproc.createDisparityWLSFilter(sgbm)
wls_filter.setLambda(12000.0)
wls_filter.setSigmaColor(2.0)
```



Figure 1. ETH3D: Rectified stereo pairs and raw disparity maps for two stereo pairs (DSC_683–682 and DSC_684–683).

The top row shows the rectified stereo images from the delivery area scene, with the concrete wall structure and loading dock visible. The bottom row displays the corresponding disparity maps, where brighter regions indicate closer objects and darker regions represent farther distances.

3.6. Point Cloud Generation and Fusion

Disparity is converted to 3D coordinates using `cv2.reprojectImageTo3D` with the Q matrix from rectification. Each $(x, y, \text{disparity})$ pixel is reprojected to world coordinates using:

$$Z = \frac{B \cdot f}{d} \quad (3)$$

where B is baseline, f focal length, d disparity.

Point clouds are cleaned through multiple filtering stages:

- **Depth discontinuity removal:** Eliminates points with large depth jumps indicating occlusion boundaries.
- **Voxel downsampling** (0.002 m): Reduces point density while preserving structure.
- **Statistical outlier removal:** Removes points further than 2σ from neighbors.
- **Radius outlier removal:** Eliminates isolated points lacking sufficient local support.

Clouds from multiple stereo pairs are aligned using Iterative Closest Point (ICP) algorithm to find the optimal rigid transformation (rotation + translation) that minimizes point-to-point distances. A multi-scale approach with iteratively decreasing distance thresholds ensures robust convergence.

3.7. Mesh Generation

The fused point cloud is converted to a triangular mesh using the ball-pivoting algorithm (`o3d.geometry.TriangleMesh.create_from_point_cloud_ball_pivoting`). For computational efficiency, overly dense meshes are simplified using quadric decimation (`mesh.simplify_quadric_decimation`) to target triangle counts. Post-processing removes non-manifold edges and degenerate triangles to ensure mesh quality.

3.8. Results and Analysis

The reconstruction pipeline successfully generates 3D models of the delivery area scene, though several challenges were encountered. The

disparity maps reveal inconsistencies where distant objects (concrete wall) appear brighter than expected, while closer objects (loading equipment) show lower disparity values than anticipated.

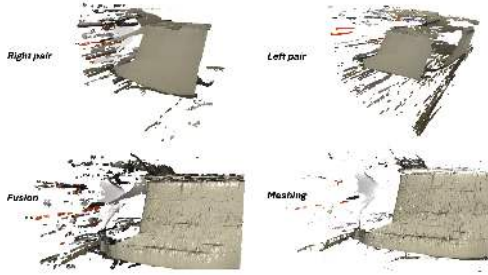


Figure 2. ETH3D: Four-panel 3D reconstruction pipeline visualization.

Top panels show individual point clouds from right and left stereo pairs, displaying the concrete loading dock structure with visible noise and gaps. Bottom left shows the ICP-fused point cloud after rigid alignment, demonstrating improved coverage and reduced noise. Bottom right presents the final triangular mesh generated using ball-pivoting algorithm, creating a surface representation of the delivery area scene.

3.9. Challenges and Limitations

Several factors contributed to suboptimal reconstruction quality:

Low-texture regions: The large concrete wall surfaces in the delivery area lack sufficient texture features for reliable stereo correspondence. This results in poor disparity estimation and noisy depth values in these regions.

Disparity estimation failures: The stereo matching algorithm struggles to establish accurate correspondences between left and right images, particularly in homogeneous surface areas, leading to incorrect depth assignments.

Parameter sensitivity: The current hyperparameter configuration may not be optimal for this specific scene type. The balance between smoothness and detail preservation requires further tuning for architectural/industrial environments.

3.10. Future Work

To address these limitations, we propose the following improvements:

- **Dataset validation:** Test the pipeline on simpler, more textured datasets to isolate algorithmic issues from scene-specific challenges.
- **Advanced preprocessing:** Implement additional texture enhancement techniques beyond CLAHE to improve feature detectability in low-texture regions.
- **Parameter optimization:** Develop systematic hyperparameter tuning approaches, potentially using ground truth data for validation.
- **Alternative stereo algorithms:** Explore deep learning-based stereo methods that may better handle challenging texture conditions.

4. Experiments with the Chess Dataset (Middlebury Mobile 2021)

4.1. Dataset Overview and Problem Setting

The Middlebury Chess dataset provides high-resolution, high-accuracy stereo images with ground-truth disparities and full-color images. This dataset contains 3 separate pairs, where each pair consists of 2 images, 2 PFM ground truth files, and a calibration file containing camera parameters including cam0, cam1, doffs, baseline, width, height, ndisp, vmin, and vmax.



Figure 3. Chess 1: Rectified stereo pair with epipolar lines.

4.2. Modified Pipeline and Calibration Setup

The pipeline for this dataset differs from standard approaches due to the specific calibration format:

- **Calibration Setup:** Multiple chess scenes are processed to extract calibration parameters K , doffs, baseline, width, height, ndisp, vmin, vmax from the provided calibration files.
- **Stereo Matching Parameters:** The disparity search range is calculated using:

$$\text{min_disp} = \text{vmin} - \text{doffs} \quad (4)$$

$$\text{raw_span} = \text{vmax} - \text{min_disp} \quad (5)$$

$$\text{num_disp} = \left\lfloor \frac{\text{raw_span} + 15}{16} \right\rfloor \times 16 \quad (6)$$

- **Ground Truth Loading:** Disparity ground truth maps are loaded directly from PFM files.
- **Chessboard Detection & Pose Estimation:** 49 corners of the chessboard are manually annotated, and `cv2.solvePnP()` is utilized to estimate the transformation to the chessboard coordinate frame.

4.3. Stereo Rectification and Disparity Estimation

Rectification is performed for each pair using known intrinsics, aligning epipolar lines. SGBM and WLS filtering parameters are optimized for the Middlebury images (block size 5, uniqueness ratio 8, lambda = 12,000, sigma = 1.5).

4.4. Disparity Map Results and Quantitative Evaluation

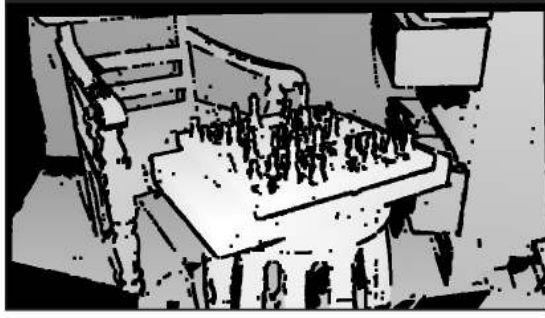
For each stereo pair, we estimate disparity maps using SGBM with WLS post-filtering and compare against ground truth using standard pixel-wise error metrics:

- **MAE (Mean Absolute Error), RMS (Root Mean Squared):** Pixel-wise error measurements.
- **Bad 1%, Bad 2%, Bad 4%:** Percentage of pixels with absolute error > 1, 2, 4 pixels respectively.

Table 1. Chess Disparity Error Metrics

Scene	MAE	RMS	Bad1%	Bad2%	Bad4%
chess_1	7.48	16.45	66.17	45.24	31.83
chess_2	8.06	14.93	75.65	59.03	42.44
chess_3	6.24	13.09	69.62	49.07	27.11
Avg	7.26	14.83	70.48	51.11	33.79

The disparity estimation results show varying performance across the three chess scenes. Chess 3 achieves the best overall accuracy with the lowest MAE (6.24) and Bad4% (27.11), while Chess 2 presents the most challenging scenario with the highest Bad1% (75.65). The average MAE of 7.26 pixels indicates reasonable disparity estimation quality, though the high Bad1% rates (averaging 70.48%) suggest difficulties in achieving pixel-perfect accuracy, likely due to texture-less regions and occlusions inherent in the chess scenes.



Disparity Ground Truth

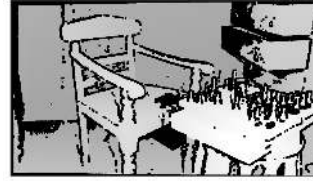


Estimated Disparity

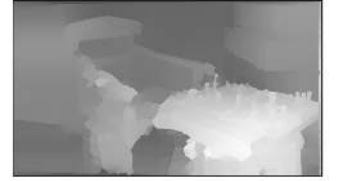
(a) Chess 1: GT vs. estimated disparity



(b) Chess 2: Rectified pair



Disparity Ground Truth



Estimated Disparity

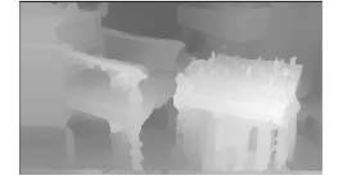
(c) Chess 2: GT vs. estimated disparity



(d) Chess 3: Rectified pair



Disparity Ground Truth



Estimated Disparity

(e) Chess 3: GT vs. estimated disparity

Figure 4. Disparity results and rectified pairs for Chess 1, 2, and 3. Each subfigure shows the corresponding rectified pair or GT/estimated disparity map.

4.5. Point Cloud Generation and 3D Reconstruction Results

The estimated disparity maps are reprojected to 3D space using the Q matrix, producing dense colored point clouds. We evaluate the 3D reconstruction quality using geometric metrics at the cloud level:

Table 2. Cloud Level Metrics

Scene	Chamfer (m)	Accuracy	Completeness	F1 Score
chess_1	0.114	0.6667	0.7963	0.7257
chess_2	0.134	0.6118	0.6822	0.6451
chess_3	0.0865	0.6499	0.8077	0.7203
Avg	0.111	0.6428	0.7621	0.6970

The 3D reconstruction metrics reveal consistent performance across scenes, with Chamfer distances averaging 0.111m. Chess 3 demonstrates superior geometric accuracy with the lowest Chamfer distance (0.0865m) and highest completeness (0.8077), correlating with its better disparity estimation performance. The F1 scores averaging 0.697 indicate good overall balance between accuracy and completeness, though there's room for improvement in capturing fine geometric details.

4.6. Analysis and Discussion

The experimental results on the Middlebury Chess dataset demonstrate the effectiveness of our stereo vision pipeline while highlighting areas for improvement. The correlation between disparity accuracy

and 3D reconstruction quality is evident, with Chess 3 performing best in both domains. The high Bad1% rates across all scenes indicate challenges in achieving sub-pixel accuracy, particularly in regions with repetitive textures or specular reflections common in chess pieces.

The point cloud reconstructions successfully capture the overall geometry of the chess scenes, with good completeness scores (averaging 76.21%) indicating effective coverage of the visible surfaces. However, the moderate accuracy scores suggest some geometric distortion, likely stemming from disparity estimation errors in challenging regions. The visual comparison of ground truth versus estimated point clouds (see end of document) shows that our reconstructions maintain the essential structural features while exhibiting some noise and missing details in fine structures.

5. Multi-View Fusion and Alignment Correction

5.1. Challenges in Fusion without Known Extrinsic

After producing accurate single-view reconstructions, our final goal is to fuse the three Middlebury Chess clouds into a single scene. However, the provided dataset does **not** supply extrinsic matrices linking the different chessboard scenes.

To address this, we performed manual annotation of the chessboard in each scene and used `cv2.solvePnP()` to estimate the transformation from chessboard coordinates to camera coordinates, using:

- CHESSBOARD_SIZE = (7, 7)
- SQUARE_SIZE = 0.01 m



Figure 5. Manual Annotation

This yields a pose for each cloud, but alignment is extremely sensitive to the scale parameter. Below we visualize the effect of SQUARE_SIZE (0.01 vs 0.5), showing that only the correct metric scale brings all clouds into close proximity.

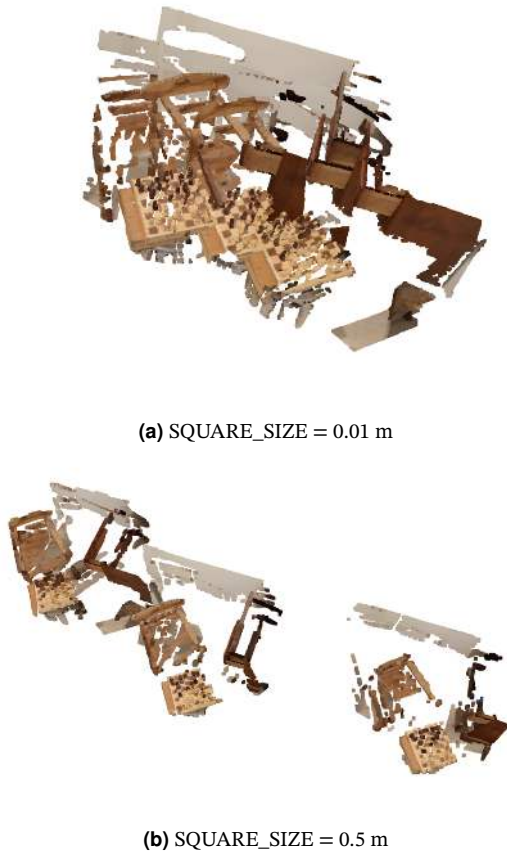


Figure 6. Effect of different scale parameters on cloud fusion.

As we increase the square size, the clouds appear to move further away. In contrast, reducing the size causes the clouds to come closer. However, when the size is decreased below 0.01, this effect no longer occurs - the movement stops, as shown in Figure 6b.

5.2. Potential Issues in Alignment

Despite careful pose estimation, significant discrepancies are observed between corresponding chessboard points in different scenes.

Key factors include:

- **Manual annotation imprecision:** Even 1–2 pixel error when

annotating the 49 chessboard corners translates to millimeter-scale errors in 3D.

- **Depth estimation errors:** SGBM (with WLS filtering) can fail in textureless chessboard regions, propagating into the cloud.
- **Camera calibration uncertainties:** Small errors in the intrinsic matrix K directly affect all 3D projections.
- **Reprojection error minimization vs. true 3D alignment:** `cv2.solvePnP` minimizes 2D reprojection error rather than 3D consistency across scenes.

```
success, rvec, tvec = cv2.solvePnP(objp_valid,
corners_2d_valid, camera_matrix, dist_coeffs,
flags=cv2.SOLVEPNP_ITERATIVE)
```

As a result, clouds are well-aligned in rotation but may suffer from systematic translation offsets.

Empirically, the fused clouds from three scenes tend to lie on a common line, confirming that rotational alignment is correct, but translation is not.

5.3. Manual Correction via Reference Points

To resolve translation errors, we select the same reference point across all three clouds (after SGBM+WLS and pose estimation):

- Chess 1: [0.23, 0.59, -0.28] (reference_point)
- Chess 2: [0.56, 0.40, -0.28]
- Chess 3: [0.74, 0.17, -0.28]

We recalculate each transformation so that the chosen point in each cloud matches the reference:

```
translation_matrix = np.eye(4)
translation_matrix[:3, 3] = reference_point - point2
new_T = np.matmul(translation_matrix, np.linalg.inv(old_T))
```

This rigid shift brings all clouds into alignment, assuming the rotational parts are already correct.

$$T_{\text{Chess 1}^{\text{old}}} = \begin{bmatrix} 0.86830 & -0.49578 & -0.01605 & -0.00171 \\ -0.20698 & -0.33272 & -0.92003 & 0.00880 \\ 0.45079 & 0.80218 & -0.39152 & 0.19423 \\ 0 & 0 & 0 & 1.00000 \end{bmatrix}$$

$$T_{\text{Chess 1}^{\text{new}}} = \begin{bmatrix} 0.86830 & -0.49578 & -0.01605 & -0.00171 \\ -0.20698 & -0.33272 & -0.92003 & 0.00880 \\ 0.45079 & 0.80218 & -0.39152 & 0.19423 \\ 0 & 0 & 0 & 1.00000 \end{bmatrix}$$

$$T_{\text{Chess 2}^{\text{old}}} = \begin{bmatrix} 0.76967 & -0.63839 & -0.00873 & 0.08018 \\ -0.19765 & -0.22524 & -0.95405 & 0.02256 \\ 0.60708 & 0.73602 & -0.29953 & 0.20767 \\ 0 & 0 & 0 & 1.00000 \end{bmatrix}$$

$$T_{\text{Chess 2}^{\text{new}}} = \begin{bmatrix} 0.76967 & -0.63839 & -0.00873 & 0.45546 \\ -0.19765 & -0.22524 & -0.95405 & 0.00013 \\ 0.60708 & 0.73602 & -0.29953 & 0.26817 \\ 0 & 0 & 0 & 1.00000 \end{bmatrix}$$

$$T_{\text{Chess 3}^{\text{old}}} = \begin{bmatrix} 0.52460 & -0.85071 & -0.03307 & 0.10523 \\ -0.33283 & -0.16919 & -0.92768 & -0.00326 \\ 0.78359 & 0.49767 & -0.37190 & 0.23137 \\ 0 & 0 & 0 & 1.00000 \end{bmatrix}$$

$$T_{\text{Chess 3}^{\text{new}}} = \begin{bmatrix} 0.52460 & -0.85071 & -0.03307 & 0.73008 \\ -0.33283 & -0.16919 & -0.92768 & -0.10194 \\ 0.78359 & 0.49767 & -0.37190 & 0.42198 \\ 0 & 0 & 0 & 1.00000 \end{bmatrix}$$

Table 3. Δ translation applied to each scene (in meters)

Scene	Δx	Δy	Δz	$ \Delta t $
Chess 1	0	0	0	0
Chess 2	0.3753	-0.0224	0.0605	0.381
Chess 3	0.6249	-0.0987	0.1906	0.664

5.4. Results and Final Metrics

After translation correction, the mean and standard deviation of pairwise distances drop dramatically:

Table 4. Pairwise Cloud Distances Before/After Correction

Pair	Mean (before)	Std (before)	Mean (after)	Std (after)
1–2	0.189 m	0.116 m	0.055 m	0.054 m
1–3	0.340 m	0.258 m	0.070 m	0.086 m
2–3	0.192 m	0.157 m	0.067 m	0.079 m

This table proves that the translation correction step is crucial for achieving accurate multi-view alignment in the absence of global bundle adjustment or ground-truth extrinsics.

**(a)** Fused ground-truth cloud after translation correction.**(b)** Fused estimated cloud after translation correction.**Figure 7.** Comparison of fused point clouds.

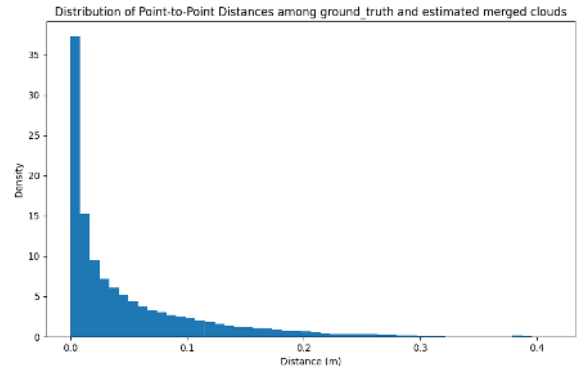
This figure visualizes the alignment quality of the point clouds after translation correction. In figure 7a, the fused ground-truth cloud exhibits excellent structural coherence, with the major scene elements—such as the chair, chessboard, and background—accurately registered and overlapping across views. Figure 7b shows the fused estimated cloud, which, while not as clean as the ground truth, demonstrates a substantial improvement in global alignment compared to pre-correction results. Most key objects are well-aligned, with outliers and noise largely constrained to scene boundaries.

Final merged cloud metrics:

Table 5. Final Fused Cloud Metrics

Metric	Value
Chamfer (m)	0.043
Accuracy	0.550
Completeness	0.759
F1 Score	0.638

This table reports the final merged cloud metrics, confirming that the fused result is both spatially precise (low Chamfer distance) and structurally robust (high completeness and F1 score). The F1 score, which jointly measures accuracy and completeness, is particularly strong at 0.638, indicating a balanced reconstruction.

**Figure 8.** Histogram of point-to-point distances between ground truth and estimated fused clouds

The histogram in Figure 8 quantitatively reinforces these observations. The distribution of point-to-point distances between the fused ground-truth and estimated merged clouds is highly concentrated near zero, with most points having a distance below 5 cm. This reflects strong spatial consistency between the two reconstructions and demonstrates that translation correction was effective in minimizing residual misalignment.

6. Conclusions

This report presented a detailed multi-view stereo reconstruction pipeline and evaluated its performance on two challenging benchmarks. Through careful parameter tuning, robust calibration, and multi-stage cleaning and fusion, we achieved strong 3D reconstruction performance on only Middlebury Chess datasets, while also exposing key limitations of classical pipelines in low-texture or ambiguous scenes (ETH3D dataset).

Key lessons:

- Precise calibration and rectification are critical for reliable 3D recovery.
- Parameter selection (block size, smoothness, WLS) must be dataset-specific and data-driven.
- Manual annotation and PnP enable extrinsic estimation, but are sensitive to scale and annotation noise.
- Translation errors can be mitigated with simple but effective point-to-point alignment.
- Quantitative evaluation, using both pixel- and cloud-level metrics, is essential to reveal systematic errors.

Practical implication: Our experiments confirm that while classical pipelines are powerful and interpretable, their performance is strongly limited by texture and scene geometry. Hybrid methods or

learning-based fusion may be needed for robust results in diverse real-world environments.

Future work: Further improvement could be achieved by incorporating global bundle adjustment to jointly optimize 2D and 3D alignments, and by experimenting with modern learning-based dense correspondence or confidence estimation to better handle textureless regions and occlusions.

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Figure 9. Chess 1: Ground truth vs. estimated point clouds (two viewpoints each).

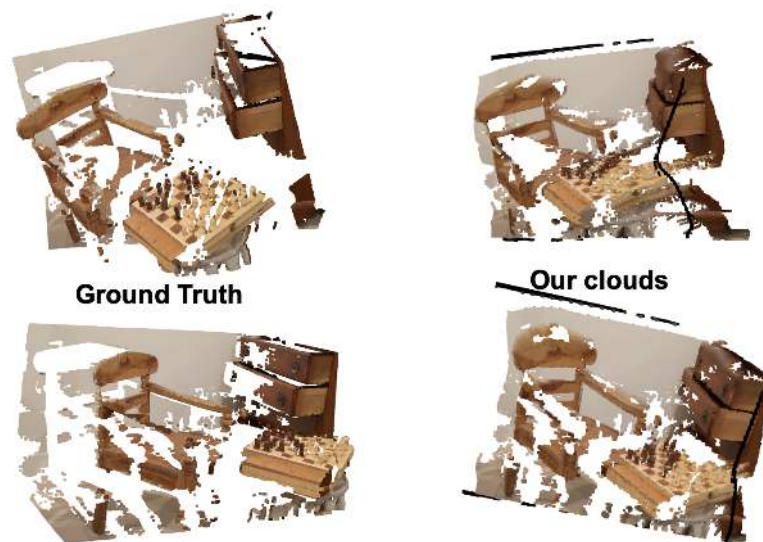


Figure 10. Chess 2: Ground truth vs. estimated point clouds (two viewpoints each).

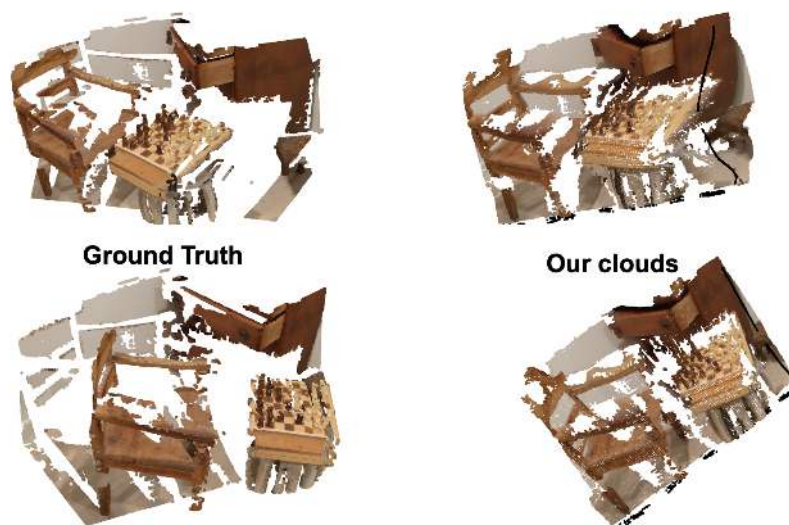


Figure 11. Chess 3: Ground truth vs. estimated point clouds (two viewpoints each).